Component Diagram - Review Document

«container»
Web Application
[React]

Runs in the browser. Shows only the sections each role can use: My Documents, Review Queue, All Documents, and Submit Document, plus the search bar, the review screen with the AI feedback, the profile, and in-system notifications. For the Super Admin: Audit Trail, User Accounts, Holiday Calendar, and AI Settings.

Sends review requests
[HTTPS / JSON]

Opens Google Workspace links in a new tab
[HTTPS]

Backend Application [Laravel / PHP (Apache application server)]
[container]

«component»
Access Middleware
[Laravel Middleware]

Checks the active session and ends it after 60 minutes of no activity or when the account is disabled. Allows only the sections, actions, and documents permitted by the role, ownership, or reviewer assignment.

Passes allowed requests (assigned reviewer only)

«component»
Review Controller
[Laravel Controller]

Handles Review Document, including View Document. Opens the assigned document and marks it Ongoing, and returns its details, history, duplicate warning, AI feedback for the reviewer's level, and the next-reviewer list. Requires the official remarks and a response to every AI suggestion, then processes Forward, Endorse, Approve, or Return.

«component»
Overdue Check
[Laravel Scheduled Command]

Runs every day. Finds reviews past five working days, marks them as overdue, and sends one reminder to the assigned reviewer and the submitter.

Checks sharing with the next reviewer (submitter's Drive permission); reads the last modified time

Opens or downloads the attached files

Records Forward, Endorse, Approve, Return, and file downloads

Notifies the next reviewer and the submitter; on Return, also the earlier reviewers

Sends the overdue reminder

Saves the review; forwards, endorses, approves, or returns

Sends the AI responses and marked remarks

Finds reviews past five working days

Checks the account's role and status
[SQL]

«component»
Google Workspace Client
[Laravel Socialite / Google API Client]

Connects to Google Workspace for Sign in with Google, link sharing checks, and reading linked documents and their last modified time. Keeps the Google Drive permission of the document roles encrypted.

«component»
Document File Service
[Laravel Service]

Saves uploaded files to private storage, and opens or downloads them only after an access check.

«component»
Audit Logging Service
[Laravel Service]

Writes read-only audit records with a server-generated timestamp, including the old and new values when something is changed.

«component»
Notification Service
[Laravel Service / Queued Job]

Saves the in-system notification and sends the matching email in the background through Gmail, recording any delivery failure. Sends only the email to an account that was disabled.

«component»
Workflow & Routing Service
[Laravel Service]

Generates reference numbers, saves documents and revisions, assigns reviewers, and changes the status following the L1, L2, L3 review order. Also cancels submissions, and moves waiting documents to a replacement reviewer or the new Division Chief (L3), returning them only if no one can take them.

«component»
AI Knowledge Service
[Laravel Service]

Adds the remarks marked for AI knowledge review and the AI suggestions marked Modify or Flag to Pending AI Knowledge for the Super Admin (never for Other), and saves every AI response for the response summary.

Checks sharing; reads the last modified time
[OAuth 2.0 / Google Drive API]

Reads the uploaded files
[Laravel Filesystem]

Sends emails through Gmail
[SMTP]

Stops and starts TAT

Reads the document, AI feedback, and duplicate warning
[SQL]

Marks documents as overdue
[SQL]

«external_system»
Google Workspace

Handles Sign in with Google, checks the sharing of Google Workspace document links, and delivers email notifications through Gmail.

«container»
Uploaded File Storage
[Private Laravel file storage (application server disk)]

Keeps uploaded PDF, DOCX, and XLSX files (up to 10MB each and 100MB per submission) exactly as submitted. Files are opened or downloaded only through the backend after an access check.

«component»
TAT Service
[Laravel Service]

Starts and stops each reviewer's TAT and counts it in working days (Monday to Friday), not counting the holidays in the Holiday Calendar or the time waiting for the submitter to revise.

Writes audit records
[SQL]

Saves in-system notifications
[SQL]

Writes review records, status, and assignments
[SQL]

Saves Pending AI Knowledge and AI responses
[SQL]

Saves TAT dates; reads holidays
[SQL]

«container»
DocuFlow Database
[MySQL (database server)]

Stores user accounts and roles with the encrypted Google Drive permission, documents and their revisions, Google Workspace links, file details and locations, document hashes, review records and remarks, AI feedback, AI knowledge and settings, the holiday calendar, in-system notifications, the audit trail, and the background job queue.